

Solving the following system:

E-Vex

$$\begin{cases} \log_2(x) + \log_2(y) = 2 \\ x + y = \frac{15}{2} \end{cases}$$

$$\log_2(x) - \log_2(y) = 2 \Rightarrow \log_2(x \cdot y) = 2$$

$$\text{then: } \frac{\ln(x \cdot y)}{\ln(2)} = 2$$

$$\Rightarrow \ln(x \cdot y) = 2 \ln 2 \Rightarrow \ln(x \cdot y) = \ln 4$$

$$\text{then: } e^{\ln(x \cdot y)} = e^{\ln 4} \Rightarrow x \cdot y = 4$$

after:

$$\begin{cases} x \cdot y = 4 \\ x + y = \frac{15}{2} \end{cases} \Rightarrow y = \frac{15}{2} - x$$

$$x \left(\frac{15}{2} - x \right) = 4$$

$$\frac{15x}{2} - x^2 - 4 = 0 \xrightarrow{\times -1} x^2 - \frac{15x}{2} + 4 = 0$$

$$2 \left(x^2 - \frac{15x}{2} + 4 \right) = 0$$

$$2x^2 - 15x + 8 = 0$$

$$x = \frac{15}{4} \quad \text{and} \quad x = \frac{\sqrt{161}}{4}$$

$$\text{when } x = \frac{15}{4} :$$

$$\frac{15}{4} + y = \frac{15}{2}$$

$$y = \frac{15}{2} - \frac{15}{4}$$

$$y = 3.75$$

$$\text{when } x = \frac{\sqrt{161}}{4}$$

$$\frac{\sqrt{161}}{4} + y = \frac{15}{2}$$

$$y = \frac{15}{2} - \frac{\sqrt{161}}{4}$$

$$y = 4.32$$

$$\int : \left(\frac{15}{4}, 3.75 \right) \left(\frac{\sqrt{161}}{4}, 4.32 \right)$$

